

Yousef Ahmed Tawfik

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PROFESSIONAL SUMMARY

Artificial Intelligence & Data Science undergraduate with hands-on experience in Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, and Generative AI. Proficient in Python, SQL, PyTorch, and TensorFlow, with a strong focus on building RAG pipelines, predictive models, and cloud-connected AI systems. Passionate about research and leveraging advanced AI to solve complex, real-world problems.

EDUCATION

Arab Academy for Science, Technology and Maritime Transport (AASTMT) — Bachelor of Science in Artificial Intelligence Cairo, Egypt
2022 – Present | Cumulative GPA: 2.9 / 4.0 | Expected Graduation: 2026

PUBLICATIONS & RESEARCH

A RAG-Enhanced Pepper Humanoid Robot for Hospital Reception and Supervised Patient Support June 2026
Accepted at IEEE International Telecommunications Conference (ITC-Egypt 2026) — To be published in IEEE Xplore

- Co-authored a paper proposing a RAG architecture integrating the SoftBank Pepper robot with cloud AI services for patient support.
- Developed a framework bridging Pepper's legacy NAOqi platform with a modern Python 3 backend (Gemini 2.5 Flash, Whisper, FAISS).
- Documented key performance metrics: 9.97% Word Error Rate (WER) and a 2.23-second average response latency.

SELECTED PROJECTS

Pepper Medical Assistant Robot — *Graduation Project & IEEE Publication* Sept 2025 – Present

- Developed an AI-powered bilingual healthcare assistant robot using SoftBank Pepper for hospital reception and patient triage.
- Implemented a RAG pipeline utilizing FAISS vector search and Gemini 2.5 Flash to deliver grounded hospital knowledge base answers.
- Integrated Whisper ASR speech recognition and cloud services, bridging Pepper's legacy environment to a modern Python backend.
- Technologies:** Python, Flask, Whisper, FAISS, Gemini 2.5 Flash, RAG, NLP, Computer Vision, SQLite, Robotics.

Tabib AI — Bilingual AI Medical Assistant — *Mobile Computing Capstone Project*

- Designed and developed a full-stack, bilingual (Arabic/English) medical assistant app featuring 17 screens in Flutter.
- Integrated Gemini 2.5 Flash to enable real-time consultations and built a 3-step symptom checker that forwards context to the chatbot.
- Configured Firebase Authentication and Realtime Database for structured patient data storage and user management.
- Technologies:** Flutter, Dart, Firebase Auth, Firebase Realtime Database, Gemini API, Provider, GoRouter, Android Studio.

Skin Lesion Classifier — *AMIT Learning Final Project*

- Designed and trained a Convolutional Neural Network (CNN) on the HAM10000 dataset to automatically classify 7 categories of skin lesions.
- Built and deployed a web interface using Python and HTML, enabling users to upload dermoscopic images and receive diagnostic support.
- Technologies:** Python, TensorFlow/Keras, Jupyter Notebook, HTML, CSS.

TECHNICAL SKILLS

Programming Languages: Python, C, SQL, Dart (Flutter), HTML, CSS
AI & Data Science: Machine Learning (ML), Deep Learning (DL), NLP, Computer Vision (CV), LLMs, Data Mining, Big Data Analytics
Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, YOLO, NumPy, Pandas, OpenCV
Advanced AI & Generative AI: Generative AI, Retrieval-Augmented Generation (RAG), FAISS, Google Gemini, Whisper ASR, Prompt Engineering
Mobile Development: Flutter, Firebase (Auth, Realtime Database), Provider, GoRouter
Tools & Technologies: Git, GitHub, Android Studio, Jupyter, Google Colab, SQLite, REST APIs, Cloud Computing, HPC

CERTIFICATIONS

AMIT Learning: Master Machine Learning & AI Diploma (220 Hours, Grade: 92%)	<i>Feb 18, 2025 – Sep 25, 2025</i>
Information Technology Institute (ITI): Machine Learning Training Program (60 Hours)	<i>Aug 14, 2025 – Aug 29, 2025</i>
The British University in Egypt (BUE): Applications of Artificial Intelligence (15 Hours)	<i>Sep 6, 2025 – Sep 8, 2025</i>